Industry classification, product market competition, and firm characteristics

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Highlights

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  Different industry classifications can generate conflicting descriptions of product market competition and firm characteristics across product market competition levels.

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  The choice of industry classification and sample period are the causes of opposite empirical results that support the conflicting theories of Schumpeter.

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  Fama French 48 and eight-digit GICS are the only classifications that generate consistent performance of industry portfolios formed by production market level from 1963 to 2017.

Abstract

We examine twelve industry classifications from three classification systems: SIC (including Fama French classifications), NAICS and GICS, over the period from 1963 to 2017. The results show different classifications can generate conflicting descriptions of product market competition and firm characteristics across product market competition levels. We also demonstrate that the choice of industry classification and sample period are the causes of opposite empirical results that support the conflicting theories of Schumpeter. We find Fama French 48 and eight-digit GICS to be the only classifications that generate consistent performance of industry portfolios formed by production market level, throughout the sample period.

Introduction

We compare the patterns of firm characteristics and returns of industry concentration level sorted portfolios among three classification systems: The Standard Industry Classification (SIC), North American Industry Classification System (NAICS), and Global Industry Classification Standard (GICS). We examine this topic because of the wide number of industry classifications used in the financial literature, and the potential for conflicting results among these studies due to variance in the firm composition and characteristics of the industry groups defined by these industry classifications. Several studies have analyzed how well various industry classifications measure the co-movements of stock returns within industry groups. However, there is no study that analyzes the cross-sectional variations of firm characteristics and consistency of the performances of industry portfolios sorted by industry concentration level over time.

On the one hand, a variety of industry classifications have been used in the financial literature with little attempt at standardization to ensure research comparability. For example, Fama and French (1997) (FF hereafter) classify firms into 48 (FF48 hereafter) industries based on four-digit SIC code to estimate the industry cost of capital. Moskowitz and Grinblatt (1999) (MG hereafter) form 20 industry portfolios based on the two-digit SIC code to examine the components of the individual stock momentum anomaly. Aghion et al. (2005) use the two-digit SIC industry classification for their research in product market competition and innovation. Hou and Robinson (2006) use the three-digit SIC industry classification to measure industry concentration and examine its impact on stock returns. Their results are contradicted by Grullon et al. (2019) using three-digit NAICS classification. The issues are not only that the sample periods of these studies vary, but also that the industry classifications themselves may not be directly comparable. Moreover, there are several other alternative industry classifications based on two, three, and four-digit SIC codes, and multiple levels of industry classifications available with NAICS and GICS.[1]

On the other hand, there are many studies in the literature that compare these three classification systems in measuring stock co-movement and industry concentration level. For example, Bhojraj et al. (2003) compare the two-digit SIC, three-digit NAICS, six-digit GICS, and FF48 classifications for S&P 1500 firms from 1994 to 2001. They find that the GICS classification is significantly better at explaining stock return co-movements and cross-sectional variations, especially among large stocks. Chan et al. (2007) compare all four levels (2, 4, 6, and 8 digits) of GICS classification with the FF48 classification for stocks from 1975 to 2004. They show that FF48 and the four-digit GICS classification are comparable in measuring stock return co-movements, but the benefits from using finer levels of classification decline at the six-digit GICS. Hrazdil and Zhang (2012) show that eight-digit GICS-based measures are better proxies for industry concentration than four-digit SIC-based measures on firms in the manufacturing sector from 1985 to 2007. Similarly, Hrazdil et al. (2014), Hrazdil and Scott (2013), and Hrazdil et al. (2013) compare the same three classification systems and demonstrate the advantage of the GICS classification in terms of grouping stocks with similar operating characteristics using all stocks in around 20-year periods from 1987–2007 and 1990–2009, respectively.

Unlike previous studies, we do not focus on the similarity (homogeneity) of firm characteristics within the industry groups. Instead, we focus on the cross-sectional variations in firm characteristics and the performance of industry portfolios sorted by product market competition level. Our motivation is to find evidence that explains the conflicting theories developed by Schumpeter in 1912 and 1942. His creative destruction theory (1912) indicates that firms in competitive industries have more incentives to engage in innovation activities. Results from Hou and Robinson's (2006) and Gu's (2016) study support this theory. Both studies use three-digit SIC industry classification. However, Schumpeter's (1942) market power theory argues that firms in the more concentrated industries have stronger incentives to innovate since they can enjoy more economic profits from their innovation. Aghion et al. (2005) and Grullon et al. (2019) studies support this theory. Gallagher et al. (2015) find strong evidence for market power theory in the Australian stock market as well. In our study, we find evidence supporting both theories. Specifically, we find that the causes are different industry classifications and choice of sample period related to stock market evolution.

The number of industries reach its peak around 1997 for the majority of the twelve classifications we study. By examining the consistency of performance before and after 1997, we find that FF 48 and eight-digit GICS are the only classifications that can produce consistent industry portfolio performance during both industry expansion (1963–1997) and industry consolidation (1998–2017).

Section snippets

Data description and the construction of different industry classifications

Our sample contains 144,387 observations on publicly held firms listed on the NYSE, AMEX, or NASDAQ from 1963 to 2017.[2] We obtain monthly share prices from CRSP and annual firm fundamentals from Compustat. Firms must have positive market equity, total assets, book equity, and at least

Industry concentration and characteristics of concentration level sorted portfolios

We measure the product market competition by the industry concentration level using Herfindahl–Hirschman Index (HHI) based on net sales[4]:

In every portfolio year, we calculate the industry concentration index by calculating the sum of all sales in every industry for each industry classification. The market share of each firm in the industry that it

Conclusion

Examining twelve industry classifications derived from three systems, we find substantial variations in the distributions of key firm characteristics such as size and scaled R&D expense when firms are grouped by concentration levels of the industries they belong to. Among them, we find support for competing theories of market structure. For example, the distributions for R&D/Assets from SIC2 and SIC3, along with all three levels of NAICS classifications, support the creative destruction theory,

Declaration of Competing Interest

None

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  After merging all the variables, firms with missing or zero dollar value in total assets are excluded. Forty-eight industries are classified using the Standard Industrial Classification (SIC) code, following Fama and French's definition as in Li et al. (2020). We define an industry dummy variable based on the NBER recession indicator (DV = 1 if NBER indicator equals 1; 0 otherwise), retrieved from the FRED database (https://fred.stlouisfed.org/series/USRECD#0).

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These 11 sectors include: Energy, Materials, Industrials, Consumer Discretionary, Consumer Staples, Health Care, Financials, Information Technology, Telecommunication Services, Utilities, and Real Estate. Note that the level 1 industry classification in WICS is the same as in GICS, which is widely used in empirical studies of sectoral analysis in international financial markets (Hrazdil et al., 2014; Chung et al., 2015; Fan et al., 2016; Li et al., 2020). Our sample covers all trading days during the fast half year of 2020, when COVID-19 rapidly broke out around the world.

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